LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Quick Revision: Sound

Key terms

Term	Meaning
Longitudinal wave	Particles vibrate along the direction of the wave. Sound in air is longitudinal.
Compression / rarefaction	Regions of high / low pressure and density
Wavelength (λ)	Distance between two consecutive compressions (or rarefactions), in m
Frequency (ν)	Number of oscillations per second, in hertz (Hz). T = 1/ν
Amplitude	Maximum disturbance; decides **loudness**
Pitch	Decided by **frequency**: higher frequency, higher pitch

v = λν · T = 1/ν · Echo: distance to the reflector = (v × t)/2
Audible range: 20 Hz to 20 kHz · Below 20 Hz: infrasound · Above 20 kHz: ultrasound

Must-know points

- Sound needs a medium; it cannot travel through a vacuum.
- Speed of sound: **solids > liquids > gases**. In air it is about 344 m/s at 22 °C, and it increases with temperature.
- An echo is heard only if the reflected sound arrives at least 0.1 s later, so the obstacle must be at least **17.2 m** away.
- Reverberation (repeated reflections) is reduced in halls with soft, sound-absorbing materials on walls and seats.
- Uses of ultrasound: detecting cracks in metal, cleaning hard-to-reach parts, scanning internal organs (ultrasonography), SONAR.

Common mistakes

- Forgetting to halve the distance in echo and SONAR questions (the sound goes there and back).
- Converting kHz: 1 kHz = 1000 Hz.

Quick check

1. A sound wave has a frequency of 500 Hz and a wavelength of 0.68 m. Find its speed.

2. A girl claps 170 m from a cliff. If sound travels at 340 m/s, after how long does she hear the echo?